

Spring 2025

Math F251X

Calculus I: Final Exam

Name: _____

Section: ☐ 9:15 (James Gossell)

☐ 11:45 (Kevin Meek)

☐ Online (Deven Barnett)

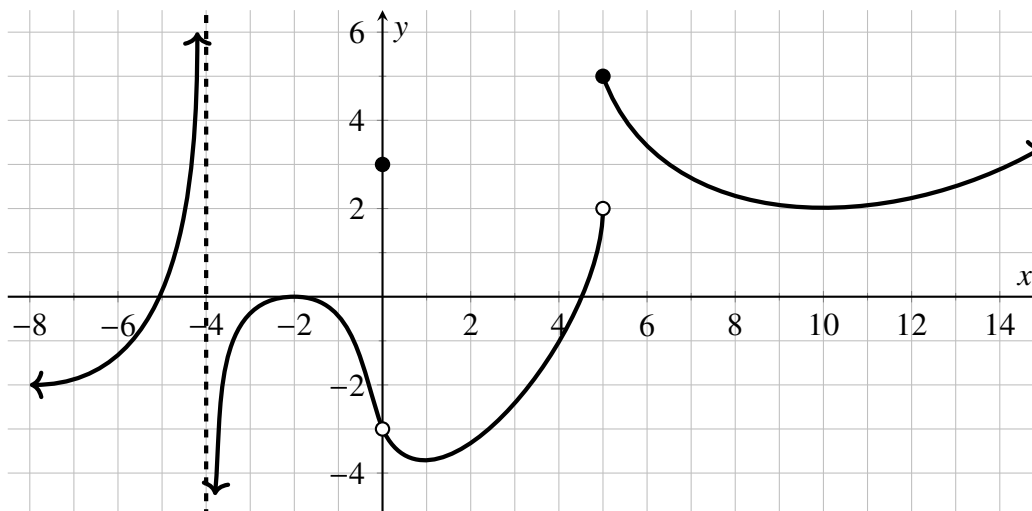
Rules:

- Partial credit will be awarded, but you must **show your work**.
- You may have a single handwritten $3'' \times 5''$ notecard, both sides. Calculators are **not allowed**.
- Place a box around your FINAL ANSWER to each question where appropriate.
- Turn off anything that might go beep during the exam.
- You have two hours to complete the exam.

Problem	Possible	Score
1	15	
2	6	
3	9	
4	10	
5	5	
6	10	
7	12	
8	12	
9	15	
10	6	
Extra Credit	(5)	
Total	100	

1. (15 points)

Consider the graph of the function $f(x)$ shown below, and answer the following questions. You must give the most complete answer; if the value is infinite, write $+\infty$ or $-\infty$.



a. $\lim_{x \rightarrow -4^+} f(x) = \underline{\hspace{2cm}}$ b. $\lim_{x \rightarrow -2} f(x) = \underline{\hspace{2cm}}$ c. $\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}}$ d. $\lim_{x \rightarrow 5^-} f(x) = \underline{\hspace{2cm}}$

e. At what x -value(s) is the function $f(x)$ **discontinuous**? _____

f. At what x -value(s) does $f(x)$ have a **local minimum**? _____

g. The line $y = -2$ is a horizontal asymptote of $f(x)$. Fill in a statement about a limit that corresponds to this fact.

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

h. On what interval(s) is $f(x)$ **concave up**? _____

2. (6 points)

Compute the following **derivatives**. Show your work clearly. You do **not** need to simplify your answers.

a. $f(x) = 3x^4 \cos\left(\frac{x^2}{5}\right)$

$$f'(x) =$$

b. $h(x) = 5 \ln(2) + \frac{x^{2.5}}{\ln(4x^3)}$

$$h'(x) =$$

3. (9 points)

Find the **absolute maximum** and the **absolute minimum** of the function $f(x) = x^3 - 3x^2$ on the interval $[-1, 4]$. Justify your answer.

maximum value of $f(x)$: _____

minimum value of $f(x)$: _____

4. (10 points)

A cabbage is launched upward at 40 meters per second from a cannon that sits 100m above the ground. The height (in meters) of the cabbage relative to the ground after t seconds is given by the function

$$h(t) = -5t^2 + 40t + 100$$

- a. Determine the velocity of the cabbage after 0.5 seconds. Include units.
- b. Determine the acceleration of the cabbage at 0.5 seconds. Include units.
- c. Determine when the cabbage will reach its maximum height. Justify your answer using calculus. Include units.

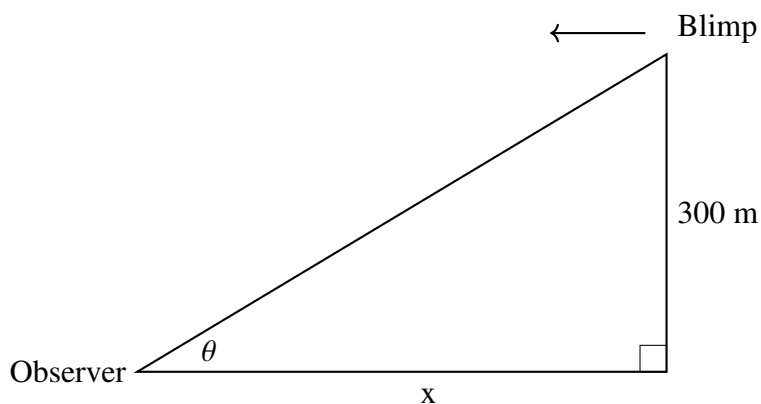
5. (5 points)

Let $h(x) = 1 - 3 \tan(x)$

- a. Find the linearization $L(x)$ of $h(x)$ near $x = 0$.
- b. Use your answer from part (a) to approximate $h(0.1)$. **Simplify** your answer.

6. (10 points)

A blimp is flying in a straight line at a constant altitude of 300 meters towards an observer who is watching from the ground. The blimp is moving horizontally at 40 meters per second. Let θ be the angle between the ground and the line of sight from the person to the blimp. (See diagram.)



- a. Write an equation that relates the variables x and θ .

- b. Determine how fast is the angle of elevation θ changing when the blimp is 400 meters horizontally from the observer? Include units in your answer.

7. (12 points)

Compute the following **limits**. Show your work clearly. Be sure to use **proper notation** where appropriate. Solutions without proper notation will not receive full credit. Indicate any use of L'Hôpital's Rule with $\stackrel{h}{=}$ or $\stackrel{L'H}{=}$ or something similar. (Be sure to verify explicitly that L'Hôpital's Rule applies!).

a. $\lim_{x \rightarrow \infty} \frac{4x}{\sqrt{x^2 - 1}}$

b. $\lim_{x \rightarrow \pi} \frac{x - \pi}{\sin(x)}$

c. $\lim_{x \rightarrow \infty} x^2 e^{-x}$

d. $\lim_{x \rightarrow 0} \frac{e^x}{\cos(x)}$

8. (12 points)

Compute the following **integrals and antiderivatives**. Give the most general answer, and show your work. Clearly indicate any substitutions you use.

a. $\int_0^3 t^2 + 6t - e^t dt$

b. $\int \csc(3\theta - \pi/2) \cot(3\theta - \pi/2) d\theta$

c. $\int \frac{12e^x}{(e^x + 12)^3} dx$

9. (15 points)

At 2:00 pm ($t = 0$), the main water line in your apartment bursts. The rate at which water is flowing from the break is given by $r(t) = \frac{5t}{1+t^2}$, where r is measured in gallons per hour and t is measured in hours.

- a. Find $r(3)$. Write a sentence that someone who has not taken calculus could understand to explain the meaning of $r(3)$. Include units.

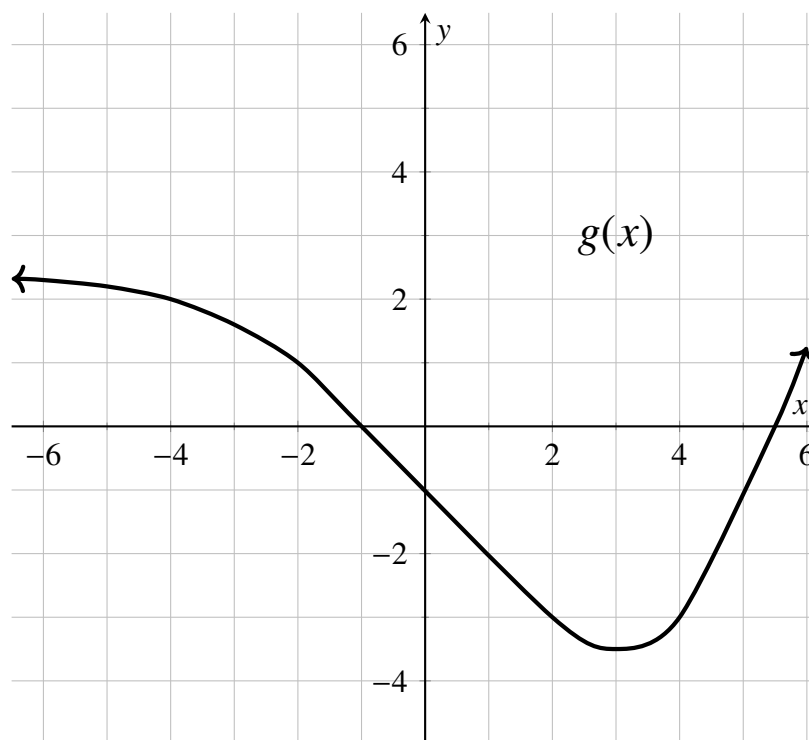
b. Evaluate $\int_1^2 r(t) dt$.

- c. Write a sentence that someone who has not taken calculus could understand to explain the meaning of $\int_1^2 r(t) dt$. Include units.

- d. At 3:00pm, is the rate of flow increasing, decreasing or staying the same? Explain how you know.

10. (6 points)

The graph of a function $g(x)$ is shown below.

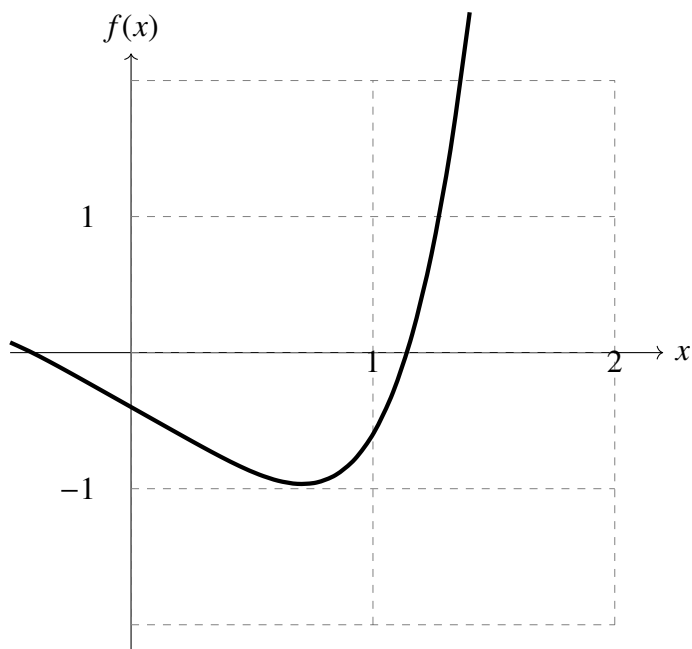


Define a new function $A(x) = \int_{-4}^x g(t) \, dt$.

a. Determine $A'(2)$.

b. Does $A(x)$ have a local maximum or a local minimum on the interval $(-4, 4)$? If so, state the corresponding x -value and explain your answer. If not, explain why not.

Extra Credit (5 points) A portion of the graph of the function $f(x) = \frac{4}{5}x^5 - x - \frac{2}{5}$ is shown below.



- a. Suppose Newton's method is used to find an approximate solution to $f(x) = 0$ from an initial guess of $x_0 = 1$. **Sketch** on the graph how the next approximation x_1 will be found, **labeling** its location on the x -axis.
- b. If your starting guess is $x_0 = 1$, **compute** x_1 .